

WolframAlpha, anyone?

Is WolframAlpha really a search engine? It does not search the web for answers, but its own database of information, and computes answers from its data.

4,609,984

The number of web sites the open directory project currently catalogues and organises into over 5,90,000 categories.

Feature

pocket, but sufficiently detailed that you can find the places that you need.

The engineer's approach to information produces too much unnecessary data when a search is performed. Data-management experience is needed to filter the data appropriately. This is the publishing approach. Information is not normally the domain of engineers, and yet it is they that have been designing search engines up until now.

The implied point is this: does having several tens of thousands of search results which you will never look at really help you answer whatever question you are posing or problem you are trying to solve? Of course not, it is just a big number that might well be impressive, but where in that haystack is the one needle that you need?

Early search engines were blind in their search methods, and it is for this reason that users needed to develop skill in entering their search in the most effective way to narrow down the results and focus in on the information they were trying to find.

Lotito gave me the example of a warehouse. If you simply fill a warehouse with boxes, then when something is needed, you will have to go through and search in all of them to find the one that is needed. But if when filling the place up with boxes in the first place you label and tag them all properly, when one is needed, you can go straight to it. It is the same difference with the information on the internet – access to this information needs to be



Facility is more flexible in its approach while searching

any way that they want. He told me "This would be very confusing. If you do not regulate this, or you do not have technology that is able to recognise when words and phrases are semantically the same.

"When you tag information, the context cannot be constructed automatically. For example, sometimes the term environment

the semantic confusion in the Ballmer example just given. In a dictionary, you might find the words "fly", "flying" and "flown" all grouped under the lemma "fly (v)", the headword or keyword for that semantic group.

Ontological search is similar to some of the principles used in Wolfram |Alpha,

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structured properly.

So, what is the approach offered by Facility? It is not a blind crawler-based search like Google, and it is not a limited computational engine such as Wolfram |Alpha. It can in fact use methods similar to both of these, and perhaps in that flexibility lies one of its strengths.

During our meeting, to stress the importance of the semantic approach in search, Lotito ran three Google searches, on "Steve Ballmer", "Steven A. Ballmer" and "Steven Anthony Ballmer". All came up with different results, and yet all refer to the current president of Microsoft. Lotito's point was that all three are semantically equivalent, and for the search to be truly meaningful, this should be understood by the engine, so that all three searches produce the same set of results.

He does not like the idea that we should simply leave people to tag information in

might refer to technology and the Windows environment; sometimes to ecology, and other times to a situation. Tagging words in this way has a level of precision that is very low. There are two ways to solve this problem: humans will tag information, or you will have an engine that will help humans to tag information. But I don't think it is possible to have a completely automatic system."

The point was made that if you had software that could accurately tag all textual information, then the tagging would no longer be needed, as the software is effectively understanding the main meanings of the words and their contexts. We are probably many decades away from this possibility, and so some kind of human assisted tagging is needed.

The approach taken by Facility involves tagging, but also the use of lemmas and ontological search. Lemmas help to alleviate

where an answer, or result, can be computed from existing and well-structured data. For example, you may have two simple taxonomies: a list of cities in England, and a list of European cities that are capitals of their countries. If you ask the question "Which is the capital city of England?" the result can easily be computed from these two lists, as London is on both lists. If you ask for the capital of England, you are asking an ontological question, because the ontology is the connection between any two taxonomies. You are asking for the intersect between these two: London is a city in England; London is a capital city; therefore London is the answer to the question "Which is the capital of England?"

The Facility engine does not avoid the algorithmic approach – it can go out there and crawl websites with all the others. It combines these different methods in a flexible manner.